Measurement 217 (2023) 112955

Contents lists available at ScienceDirect

Measurement

journal homepage: www.elsevier.com/locate/measurement

Developing a digital twin model for monitoring building structural health by combining a building information model and a real-scene 3D model

Jinghai Xu [a,*], Xin Shu [a], Peng Qiao [b], Shanshan Li [a], Jigang Xu [c]

[a] School of Geomatics Science and Engineering, Nanjing Tech University, 30 South Puzhu road, Nanjing, China
[b] Science and Technology Development Center, Jiangsu Provincial Department of Housing and Urban-Rural Development, 287 North Jiangdong road, Nanjing, China
[c] School of Civil Engineering, Nanjing Tech University, 30 South Puzhu road, Nanjing, China

ARTICLE INFO

Keywords:
Real-scene 3D model
Building information model (BIM)
Digital twin
Oblique photogrammetry
Structural health monitoring (SHM)

ABSTRACT

Structural health monitoring (SHM) is a critical component of ensuring the safety of buildings, yet the effective management of sensors and their corresponding monitoring data is complex. Thus, a digital twin health monitoring information model for buildings was proposed to simultaneously locate sensors in space and monitor data. We described the requirements of this process and designed a conceptual model by combining a building information model (BIM) and a real-scene 3D model. The key steps in the fusion process include reconstructing the BIM into sub-BIMs considering building component types and geometric transformations, 3D spatial position registration with a rigid-body transformation method, and semantic information mapping. After the fusion process, monitoring data from sensors are stored in a database management system and integrated with the fused model. A digital twin model of the Nanjing Museum Old Hall was constructed as a case study, which evaluated the fusion method's feasibility and usability.

1. Introduction

Ensuring the safety of buildings is a critical component of urban disaster prevention. In addition to external factors, such as natural disasters, the safety of buildings can also be affected by internal factors such as material performance degradation and component creeps [39]. However, structural health is dynamic, and when the health status of a building varies greatly from the design expectations, accidents may result. For example, the Jiaxin Hotel in Quanzhou, China unexpectedly collapsed in March 2020, causing 29 deaths and 42 injuries during its reconstruction[1]. In another incident, 53 people died when an old building in Changsha City was suddenly damaged and collapsed in April 2022.[2] To prevent further loss of life owing to lapses in building safety, ongoing monitoring is required. Thus, structural health monitoring (SHM) technology has been widely used to collect and analyze data to diagnose structural health status using non-destructive sensors arranged inside the structure [22].

Early studies and applications of SHM in civil engineering focused on projects related to water conservation, such as bridges and dams [18]. However, with the increase in the construction of large or structurally complex buildings, SHM technology has been gradually introduced into the field of building engineering. Multiple aspects of SHM technology have been investigated, such as high-performance sensors, data collection, automatic monitoring systems, and methods of identifying structural damage [4,15,36].

However, accurately evaluating and predicting a building's health state based on SHM data is challenging owing to the diversity in the structural forms of building engineering, the complexity of the internal structural components (e.g., beams, plates, and columns), and the substantial variation in the states of these two factors [31]. To address this issue, numerous studies have focused on improving the methods of SHM data analysis. For example, to prevent the impact of anomalous monitoring data, Zhang et al. [38] described a simple and efficient approach that incorporates complementary information obtained from multi-view local binary patterns and the random forest algorithm to distinguish data anomalies. Additionally, Jayawickrem et al. (2022) reviewed the

* Corresponding author.
E-mail address: xu_jing_hai@163.com (J. Xu).

[1] https://baike.baidu.com/item/3%C2%B77%E6%B3%89%E5%B7%9E%E6%AC%A3%E4%BD%B3%E9%85%92%E5%BA%97%E5%9D%8D%E5%A1%8C%E4%BA%8B%E6%95%85/24517017?fr=aladdin.

[2] https://baike.baidu.com/item/4%C2%B729%E9%95%BF%E6%B2%99%E8%87%AA%E5%BB%BA%E6%88%BF%E5%80%92%E5%A1%8C%E4%BA%8B%E6%95%85/60965288?fr=aladdin.

https://doi.org/10.1016/j.measurement.2023.112955
Received 29 November 2022; Received in revised form 14 April 2023; Accepted 25 April 2023
Available online 3 May 2023
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application of optical fiber sensor technology and deep learning approaches in the data processing capability of SHM technology. Related data-processing methods, including Bayesian inference method, the Kalman filter algorithm, fuzzy logic, and various machine learning approaches, have also been explored [24,13,21].

In addition to the data-processing methods, data models are an important component of addressing the complexity of building safety. For example, the structural mechanics model supports the analysis of structural health using the finite-element method. In this study, a new building SHM information model based on digital twin technology was proposed to extend and complement the structural mechanics model. The objectives of this study were to: (1) manage the monitoring sensors located on the internal components of a building and their monitoring data in space and (2) analyze the external influences on the building. Subsequently, the association of the monitoring data of different structural components as a whole in real time helps restore the dynamic health state of the building. Furthermore, the external environment is another crucial factor that can cause damage to buildings. Thus, the second objective of our study was to analyze the external influences on the building. The model offers a foundation for monitoring data management, analysis, and the dynamic representation of the structural health state.

Digital twin technology refers to the process of mapping an object in the real world to create a highly accurate digital representation and reflecting the information collected in the real world onto the digital model to make predictions based on the simulation, analysis, and visualization of the natural world [27]. A digital twin maps the attributes, structures, states, performances, and functions of physical entities and systems within a virtual world [6]. Grigor et al. [14] addressed the methodological development of structural simulation, analysis, and control models for historic buildings using the concept of digital twin technology and demonstrated that developing a digital twin model for a historical building allows for more extensive usage of SHM data. Additionally, these models facilitate a greater understanding of the real-time structural behavior of complex buildings to ensure general structural safety with minimal technical interventions. Jiang et al. [17] introduced the concept of a digital twin within the field of bridge engineering and established a general, non-deterministic, digital twin-driven multiscale fatigue life prediction framework for steel bridges, which enhanced the understanding of the fatigue evolution mechanism throughout the bridge lifecycle.

The direct exploration of establishing a digital twin model for building an SHM system is limited and requires further investigation. However, studies on the potential applications of the digital twin concept in evaluating smart cities may provide a point of reference. Batty [3] analyzed the construction methods of digital twin cities, including that of high- and low-frequency cities. Similarly, White et al. [30] analyzed the modeling and construction of a digital twin smart city and found that it is possible to create an accurate digital twin by combining a large BIM with the big data generated by the Internet of Things (IoT) sensors. Further, Jaume et al. [16] conceptualized the use of digital twins in urban management and defined the basic structure of a city's digital twin. Additionally, Du [12] reported that creating digital twin buildings may involve constructing physical information systems that use buildings as carriers and utilize BIMs, geographic information systems (GIS), the IoT, artificial intelligence, big data, and other technologies. In comparison, Zhu et al. [40] introduced digital twin technology in the construction of the Sichuan–Tibet railway and explored the methods for classification and coding of physical elements.

As shown above, realizing a digital twin model of building health monitoring is important by combining a BIM and a real-scene 3D model. In particular, BIM technology can provide the distribution details of a building's internal components (such as beams, plates, and columns) in a 3D form and integrate information from construction to operation to provide a foundation for collecting and managing a building's health data from monitoring sensors. To this end, BIM technology has been introduced into the process of monitoring building health, including the visualization of monitoring information, comprehensive management of structure monitoring information, and establishment of a structural safety early warning system [25,23,19,29]. For example, Boddupalli et al. [5] adopted a BIM as a health monitoring computing and digital platform to organize and visualize health monitoring data and verify the advantages of BIM technology in these capacities. Further, Yang [34] and Deng et al. [11] explored the application of BIMs in SHM visualization, while Theiler and Smarsly [28] proposed a BIM-based health monitoring information description model by extending the industry foundation class (IFC) model. This model uses IFC as a formal basis to describe the information of the SHM system and promote the development of the interoperability between the SHM systems.

Although BIMs provide the internal location information of a structure for monitoring sensors, which is conducive to analyzing the relationships among different structural components, health monitoring systems should also focus on the impact of a structure's external environment. For example, the impact of surrounding disturbances and seismic waves should be monitored. Therefore, research on the SHM of a building based on its spatial location is required.

One such tool that may be utilized for this purpose is GIS, which is essential for urban information analysis [33], emphasizes the geographical and spatial characteristics of objects, and provides accurate positioning and spatial analysis of buildings. An important development in the current GIS field is the real-scene 3D model, which results from the oblique photogrammetry method, and possesses the characteristics of high accuracy, fidelity, and measurability and has become an important technology in the construction and management of smart cities [10,35]. Recently, the Chinese government issued a notice to promote the construction of a 3D realistic geospatial scene in China, aiming to develop new 3D geospatial information products to describe and represent the real 3D spaces of industrial production, living environments, and ecological development [8]. However, the real-scene 3D model represents the "skin," or exterior, of the building without the interior component information, which inhibits its capacity to meet sophisticated needs such as indoor navigation, fire evacuation, and building construction.

In summary, it is necessary to merge the complementary real-scene 3D model and BIM to establish an accurate digital twin model of buildings for SHM. Chen et al. [9] explored feasible approaches for converting IFC models into 3D Tiles based on an open-source solution for visualizing BIM in Cesium. Following the open-source solution, we aim to merge a BIM and a real-scene 3D model to create a digital twin model for building SHM. The remainder of this paper is organized as follows: Section 2—the conceptual model of the digital twin, Section 3—the proposed methodology for merging a BIM and a real-scene 3D model, Section 4—the application of the methodology to the Nanjing Museum, and Section 5—conclusions.

2. Conceptual model of the digital twin

2.1. Analysis of model requirements

To monitor and manage a buildings' health information in real-time, its digital twin model needs to meet three requirements, as shown in Fig. 1: (1) The ability to manage structural information such as the building's size and geometric parameters, as well as semantic information, such as the materials used for beams, plates, and columns and their prefabricated assembly. The structural information of buildings can be used for dynamic and static mechanical analysis of structures under the influence of the environment. (2) The ability to describe the external shape of the building in terms of its location, contour, and geometric shape. This information can be used to analyze the relationship between the building and its surroundings, such as the impact of foundation pit excavation on surrounding buildings during the construction period or the impact of surrounding disturbances on the building. (3) The ability

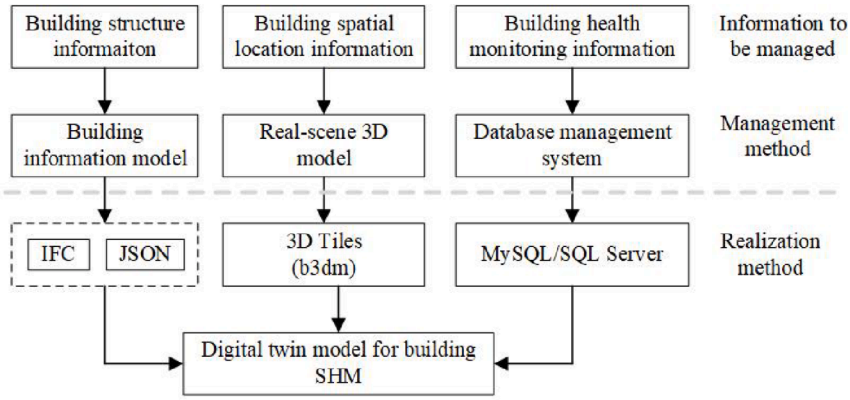


Fig. 1. Conceptual model of the digital twin model for building structural health monitoring (SHM) systems. IFC, industry foundation class; JSON, JavaScript Object Notation.

to manage the build-state monitoring information. In addition, the model should support managing dynamic monitoring information, such as stress, displacement, and vibration, from various sensors. These sensors collect real-time information on the external representation of the building structure and internal changes in the structure.

Building structure information, external shape information, and sensor data can be considered the geometric and attribute information of a GIS, and existing GIS data models can be referenced. The BIM is convenient for managing building component information and is widely used in the construction industry. However, this model can also represent and manage building information during the planning, design, construction, and maintenance periods. Therefore, we selected the BIM to represent and manage the building structure information in this study. The real-scene 3D model provides accurate geographic location information, realistic effects, and accurate measurements. Therefore, we selected a real-scene 3D model to describe the geographic information of the building. Combining a real-scene 3D model with a BIM can meet the first two requirements outlined above and provide a foundation for locating sensors and analyzing the surrounding environment. For the third requirement, the building SHM data are essentially structured data that can be managed using a database. The monitoring data were then easily associated with the geometric data of the fused model as attribute data for the GIS model. Therefore, the fusion of a BIM and a real-scene 3D model was the key to building a digital twin model in this study.

2.2. Selection of data standards for the model fusion

A basic model for converting a BIM into a real-scene 3D model is required to support fusion and combine BIMs and real-scene 3D models. For the GIS model, we selected a real-scene 3D model as the target/foundation model for fusion because it simultaneously manages the geometry and dynamic monitoring data of a building. Thus, the data must first be transferred from the BIM to a real-scene 3D model to merge the two models.

In recent years, researchers and engineers have developed many data standards for BIMs and real-scene 3D models. The optimal data standard that is a good starting point (standard of the source model) and endpoint (standard of the target model) for the fusion must be addressed. Several professional BIM software programs, and their corresponding model standards, are available including: the RVT extension of Autodesk's Revit product files, the CATIA program used by Gehry Technology, and the PLN files created with Graphisoft ArchiCAD. Furthermore, the IFC model has become a widely used BIM standard [1]. This standard was formulated and managed by the International Alliance for Interoperability and displays the characteristics of stability, openness, and integrity. Therefore, the IFC standard was selected as the source model standard for the model fusion in this study.

A real-scene 3D model is generally created using the oblique photogrammetry method to provide a near-real 3D model of physical objects. Many commonly used data standards have been produced for real-scene 3D modeling programs such as Astrium's Street Factory, photoscans, and PIX4Dmapper. The commonly used model standards for real-scene 3D models include OpenSceneGraph (OSGB), object file (OBJ), 3D Tiles, and S3c. Among them, the 3D Tiles model is an open-source standard that is easily accessible to the public and provides good support for rendering the 3D Digital Earth [37]. Therefore, the 3D Tiles model was used as the target standard for the model fusion in this study. The basic structure of the 3D Tiles standard is shown in Fig. 2. The tiles in the model store all 3D geographic information data. The batched 3D Model (b3dm) standard tiles are suitable for mass 3D model loading, support rich texture and material information, and are widely used to represent buildings. The data in the "tileset.json" file describes the tiles and the storage of the spatial organization structure of the tiles.

3. Merging a BIM with real-scene 3D model

As mentioned in the previous section, the key to model fusion is transferring the BIM from the IFC standard to the real-scene 3D model in the 3D Tiles standard. The difficulty in fusion lies in transforming the BIM's geometric, spatial, and semantic information. The fusion framework is designed as shown in Fig. 3. The BIM of buildings in the IFC standards must undergo a series of transformation processes, which mainly include four steps: (1) BIM reconstruction, (2) geometric information conversion, (3) spatial position registration, and (4) semantic mapping of the construction information. During the BIM reconstruction step, the model must be split into sub-BIMs according to their component types. Subsequently, their geometric and semantic information is obtained, which lays the foundation for geometric and semantic information transformation. The geometric information conversion step

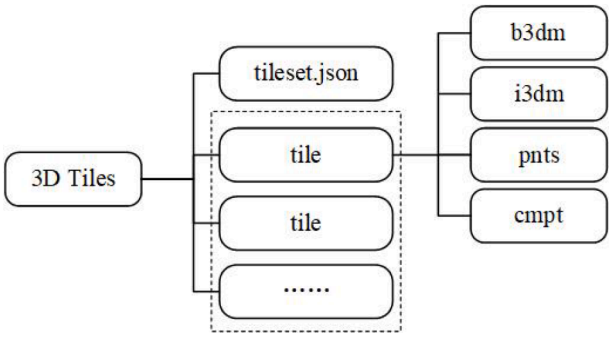


Fig. 2. Structure of 3D Tiles format. b3dm, batched 3D model; i3dm, instanced 3D model; pnts, Point Cloud tile; cmpt, Composite tile.

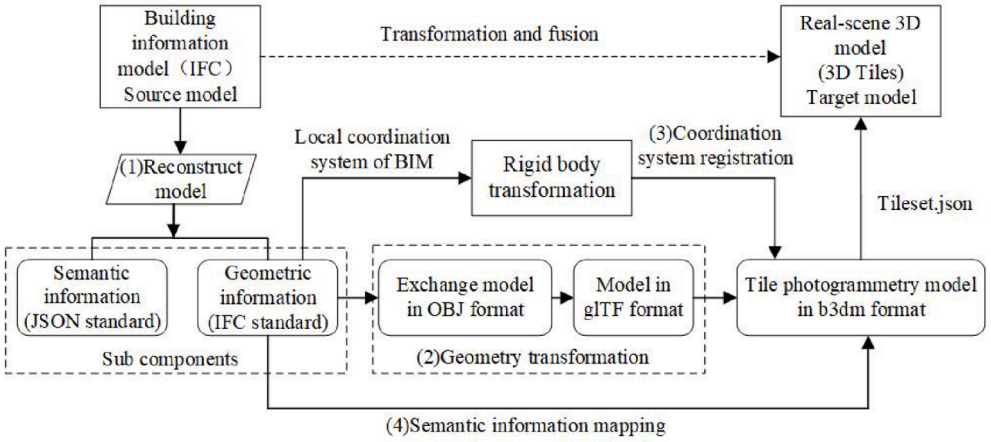


Fig. 3. Framework of the fusion of building information model (BIM) and real-scene 3D model. IFC, industry foundation class; OBJ, object file; JSON, JavaScript Object Notation; glTF, GL transmission format; b3dm, batched 3D model.

transforms the BIM in the IFC standard into a 3D Tiles standard; thus, the geometric information is merged. Although BIM and real-scene 3D models have different coordination systems, the spatial position registration step solves this problem. The semantic information of the BIM is converted and merged into the corresponding real-scene 3D model through semantic mapping of construction information.

3.1. BIM reconstruction

A BIM contains geometric and semantic information for several building components, such as the size of walls and the specifications for column materials. A BIM can easily lose semantic building information when directly converted into a model with a 3D Tiles standard. To avoid this, we first deconstruct the BIM and then reconstruct it into a set of BIMs with multiple sub-components. Each sub-component model uses the IFC standard to store geometric information, such as beam1.ifc, beam2.ifc, and column1.ifc. These sub-components store the corresponding semantic attribute information in an attribute file with the same name. The attribute file is stored in JavaScript Object Notation (JSON) format, which is a lightweight data-exchange format suitable for transmitting and exchanging text data [7]. The BIM reconstruction process is illustrated in Fig. 4.

In this process, accurately splitting the BIM, forming sub-components, and accurately linking and saving the geometric and attribute information are crucial for reconstruction. Building components were classified based on their type, including building columns, structural columns, walls (such as internal partitions and curtain walls), stair railings, ramps, doors, and windows. A few software packages can be used for reconstruction operations, such as a secondary development method based on Autodesk Revit. Considering software copyright, an open-source software BIM server was used to reconstruct the BIM. As a commonly used open-source software in the BIM field, the BIM server provides IFC file management, data parsing, and formatted output and can directly open the model in the RVT standard. With the support of the application programming interface (API) of the BIM server, we developed a program to implement a series of reconstruction processes, such as sub-component traversal, component type discrimination, semantic information reading, and component data export.

The type of each sub-component is queried by traversing all sub-components in the BIM. The following types of sub-components are available: stair railings, doors, and windows. These sub-components are identified as the ones to be reconstructed; their attribute information will be explored further. The reading and parsing rules vary depending on the geometric characteristics of components, which are mainly divided into three categories: (1) A simple component—conventional read-and-parse statements of the BIM server are used to obtain the component geometric and semantic information, such as windows, doors, and columns. (2) Components with holes—in addition to reading and parsing the normal information of the component, nested hole information, such as walls, must be read and parsed. (3) Objects containing composite components—in addition to reading and parsing the components' normal information, the composited components' subparts need to be read and parsed according to associative properties, such as stairs containing railings, stair runs, and floors.

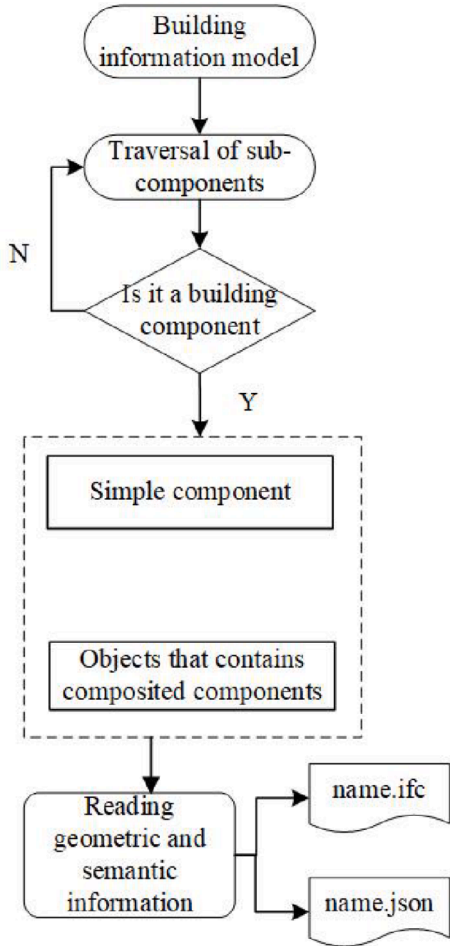


Fig. 4. Reconstruction process of BIM.

After the component information is read, it is stored in the geometric information (IFC) and semantic information (JSON) file formats. The file names for the IFC and JSON files reflected a one-to-one correspondence between the geometric and semantic information of the components.

3.2. Geometric information transformation

The geometric information in the IFC standard cannot be directly converted into a 3D Tiles standard. Therefore, an indirect conversion method is adopted. First, the OBJ format is selected as the intermediate exchange format. This object file format was a standard 3D model developed by Alias Wavefront. This standard is recorded in text and is widely used to exchange 3D software models. Subsequently, the OBJ format was converted to the GL transmission format (glTF) in the real-scene 3D model. The glTF format was converted to the b3dm format. The tile file in the 3D Tiles was established to realize the geometric information conversion of BIM using the b3dm file.

During the geometric transformation process, an open-source tool, IfcConvert (https://ifcopenshell.org/ifcconvert.html), was used to provide the conversion function from the IFC format to the OBJ format. To fully convert geometric information from the IFC format to the b3dm format, it is necessary to focus on the transformation from the OBJ to the b3dm format. The main part of the b3dm model that describes the geometric element information is in binary glTF format, which can reduce redundant data irrelevant to rendering in a 3D format and is a 3D file format suitable for OpenGL loading. With the development of 3D technology, various 3D standards have emerged, such as Blender, Maya, and Lightwave3D. However, glTF is committed to providing an exchange format for web 3D models.

The data structure framework of the glTF format differs from that of the OBJ format. As shown in Fig. 5, the conversion of vertex coordinates, external normal vectors, texture coordinates, texture images, and other information in the OBJ standard is required. The glTF bin file stores the basic geometric data; therefore, it is necessary to save building vertex coordinates, external normal vectors, and texture coordinates in the OBJ. Texture images of buildings were stored in formats including glTF, PNG, and JPG. In OBJ, triangular patches are drawn using a vertex index to form the geometry. With glTF, a mesh of primitive arrays was used to represent a single complex geometry. Each primitive array contains the corresponding vertex attributes and textures referenced in a bin file. Therefore, setting the vertex traversal mode is vital for ensuring that the grid can be converted correctly.

Several glTF conversion tools have been developed, including the relatively sophisticated obj2gltf, an open-source library launched by Analytical Graphics, Inc. (https://github.com/AnalyticalGraphicsInc/obj2gltf). Based on the process shown in Fig. 5, the OBJ standard was converted to glTF using the tool obj2gltf.

3.3. 3D spatial position registration

The 3D Tiles standard is a GIS model that typically has a spatial coordinate system, such as the WGS84 world coordinate system. The BIM coordinate system in the IFC is typically a local coordinate system that can describe the scale and size of a building. However, it is necessary to register the spatial positions of the two models under 3D registration to correctly display the BIM in the GIS.

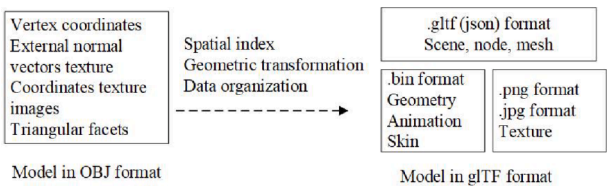


Fig. 5. Conversion from OBJ to glTF.

The source model was the BIM and the real-scene 3D model was set as the target model. The iterative closure point (ICP) algorithm is widely used, particularly in the point-cloud field, when considering multiple algorithms for registering 3D spatial models. The implementation process of the ICP algorithm involves repeatedly selecting the tie-point pairs in two point clouds and calculating the optimal rigid-body transformation until the convergence accuracy requirements for correct registration are satisfied [32,20]. The ICP algorithm is suitable for registering multi-station ground point clouds and can achieve good results. The coordinate system of BIM differs considerably from that of a real-scene 3D model. However, the ICP algorithm has weaknesses in registering BIMs, such as the slow speed of iterative convergence and low accuracy.

The BIM and real-scene 3D model have prominent rigid-body characteristics, and both models have good geometric characteristics conducive to determining the point pair relationship. Therefore, accurate rigid-body transformation parameters can be calculated according to the tie-point pairs of the building. Afterwards, their fusion transformation can be realized. The singular value decomposition (SVD) method in rigid-body transformations has the advantages of easy implementation, simple calculation, high speed, and reliable accuracy [26,2]. Therefore, this method is suitable for the registration of BIMs and real-scene 3D models.

The first step of the registration process is to choose tie points that mainly adhere to the following criteria: (1) Tie points should be selected from visible feature points located on the building surface. Since buildings in a real-scene 3D model usually have external surface information alone, it is unsuitable to use the internal points of the building as tie points. Instead, points on the building surface should be selected, such as wall corners, window corners, and building-surface structure joints, to improve the accuracy of point acquisition. (2) Tie points should be evenly distributed. The positions of the selected tie points should be reasonably distributed according to the scale of the building to avoid excessive concentrations or linear arrangements, which would affect the results. (3) There should be at least three tie points. The SVD method requires at least three groups of initial data points to determine registration parameters. The number of tie points can be appropriately increased to improve the calculation's accuracy. After the tie-point pairs are selected, the registration calculation is performed, as shown in Fig. 6. The source data is the BIM data, and the coordinates of the tie points in the BIM are expressed as $TP_i^{'} = (x_i^{'}, y_i^{'}, z_i^{'})$. The real-scene 3D model data were regarded as the target data, and the coordinates of the tie points were expressed as $TP_i = (x_i, y_i, z_i)$.

The detailed processes for generating solutions for the registration parameters are summarized as follows (Eqs. (1)–(6):

(1) Calculation of the center of gravity of two sets of coordinates.

$$P = \frac{1}{N}\sum_{i=1}^{N} TP_i Q = \frac{1}{N}\sum_{i=1}^{N} TP_i^{'} \tag{1}$$

where P represents the center of gravity of the tie points of the BIM, Q

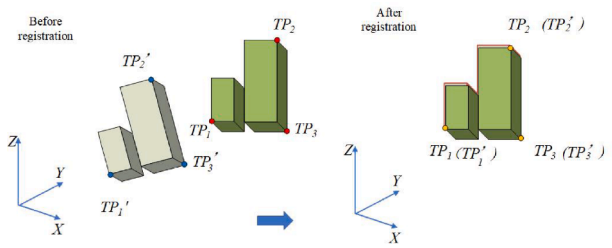


Fig. 6. Registration of building information model (BIM) and real-scene 3D model. TP_i represents coordinates of the tie point of the real-scene 3D model, and $TP_i^{'}$ represents coordinates of the tie point of the BIM.

represents the center of gravity of the tie points of a real-scene 3D model, N is the pair number of selected tie points, i is the serial number of tie points, TP_i represents coordinates of the tie point of the real-scene 3D model, and $TP_i^{'}$ represents coordinates of the tie point of the BIM.

(2) Barycentrization of model coordinates.

$$\{\Delta TP_i = TP_i - P\}\{\Delta TP_i^{'} = TP_i^{'} - Q\} \tag{2}$$

where $\Delta TP_i^{'}$ represents the coordinates of the BIM after barycentrization, and ΔTP_i represents the coordinates of the real-scene 3D model after barycentrization.

(3) Calculation of 3 × 3 transformation data matrix S.

$$S = \sum_{i=1}^{N} \Delta TP_i^{T} \Delta TP_i^{'} \tag{3}$$

where S represents the transformation matrix, which is decomposed into rotation and translation matrices for the convenience of the transformation operation.

(4) Decomposition of the matrix based on the SVD algorithm.

$$S = U\Lambda V^T \tag{4}$$

where, U and V are unitary matrices composed of left and right singular vectors, respectively, and Λ are diagonal matrices composed of singular values.

(5) Calculation of the rotation matrix R.

$$R = VU^T \tag{5}$$

(6) Calculation of the translation matrix T.

$$T = Q - RP \tag{6}$$

To complete the calculation of the rotation and translation parameters, we solved the parameter values of the SVD function using MATLAB software and completed the registration with Python.

3.4. Mapping semantic information

A BIM includes the semantic information of sub-components, such as the beam material and concrete strength. The semantic information of the sub-components was stored in the corresponding JSON files through the previous reconstruction steps. Semantic mapping aligns the JSON semantic information of multiple sub-components in the BIM to the semantic attribute information in the real-scene 3D model. However, it associates the semantic information of each sub-component with its glTF geometric information individually. Thus, the converted b3dm file was obtained and integrated into 3D Tiles. This accomplished the fusion of the BIM and real-scene 3D model.

As shown in Fig. 7, the b3dm file comprises a file header and body. The file header records the file type, version, size, and other relevant information. The file body includes a feature table, batch table, and binary glTF, which record the number of features, feature attributes, and geometric data. Therefore, the essence of semantic mapping is to place the glTF and JSON data of the sub-components into the corresponding blocks of the b3dm file and maintain their original relationships.

In the semantic mapping process, the JSON file is first generated based on the name of the glTF geometry file of each sub-component. For example, wall0.json has been indexed according to wall0.gltf, and they have a common ID. The glTF data of the sub-components, such as wall0.gltf, is binarized and written in the binary glTF data block of the b3dm file. Meanwhile, a semantic information file, such as wall0.json, corresponding to the sub-component is written into the batch table data block of the b3dm file. The identification (ID)between walls wall0.gltf and wall0.json was converted to a batch ID to maintain its connection. Finally, a tool is developed to realize the semantic mapping based on the open source 3D Tiles tool sample generator published by Cesium (https://github.com/AnalyticalGraphicsInc/3D-tiles-tools).

4. Application and discussion

4.1. Case study

The Nanjing Museum is one of the most important historical and cultural landmark buildings and the first large comprehensive museum invested by the state of China. The museum's Old Hall has a long history spanning more than 70 years. In 2008, the museum was lifted through raising columns 3.0 m after cutting on the first floor. In addition, isolation bearings were installed at the base of the building for seismic reinforcement. With an increase in service time, the stiffness of the rubber isolation bearing is prone to increase, and the horizontal deformation capacity decreases. A deterioration in the mechanical properties of isolation bearings affects the safety of isolation structures under strong earthquakes. Portions of the Old Hall's structural components may be damaged by nature or humans, which in turn will impact the safety of the entire structure. Therefore, it is necessary to monitor the structural health of the Old Hall. To this end, 60 sensors were deployed for vibration isolation on nine abutments (A–I), as shown in Fig. 8 and Table 1. Illustrations of the sensors are displayed in Fig. 9.

We developed a digital twin model for the Old Hall SHM system, based on the combination of a BIM with a real-scene 3D model, and the integration of sensor monitoring data. Tilt photos were acquired using a DJI Matrix 600 electric multirotor unpiloted aerial vehicle (UAV) with a five-lens tilt camera. The parameters of the UAV platform and cameras used in the aerial survey are listed in Table 2. During the aerial survey, the azimuth overlap was 80 % and the lateral overlap was 75 %. As a result, more than 1600 aerial photographs were obtained, and the data volume was approximately 11 gigabytes. After post-processing these photos, the real-scene 3D model of the Nanjing Museum in 3D Tiles standard was established through context capture smart 3D software, as shown in Fig. 13.

The BIM of the Old Hall of the Nanjing Museum was manually modeled using Revit software. The BIM of the Old Hall was first reconstructed using the open-source software BIM server (Fig. 10) to

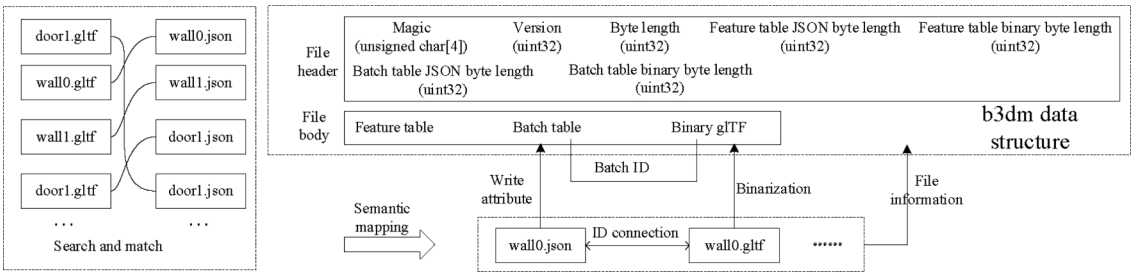


Fig. 7. The process of mapping semantic information. JSON, JavaScript Object Notation; glTF, GL transmission format; b3dm, batched 3D model.

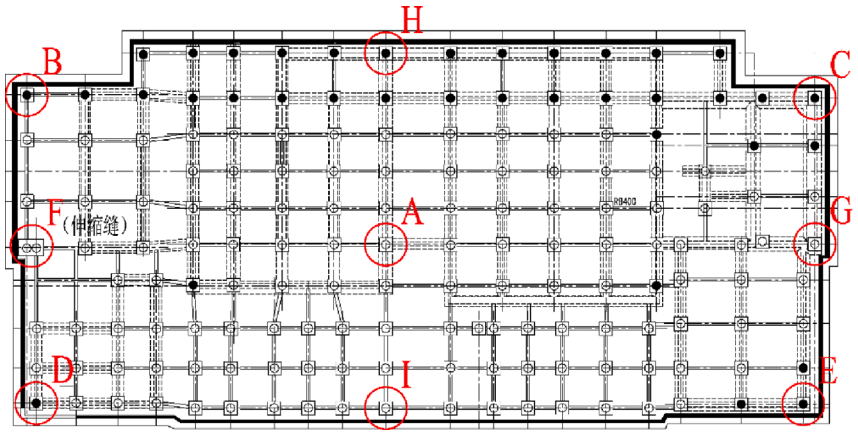


Fig. 8. Plane distribution of sensors in the isolation storey.

Table 1
Sensor information of the Nanjing Museum Old Hall.

Sensor type	Number	Sampling frequency	Location
Vibration sensor	21	20 HZ	A–H
Displacement sensor (computer vision)	9	10 Hz	A–H
Displacement sensor (optical fiber)	10	20 Hz	A–H
Strain sensor	10	20 Hz	A–H
Temperature	10	1 Hz	A–H

accomplish the fusion of the BIM and the real-scene 3D model. Using the operating API functions of the BIM server, the BIM of the Old Hall was reconstructed as a series of sub-BIMs with sub-components. The new sub-BIMs are in the IFC and JSON formats (as shown in Table 3). The sub-BIMs are registered with a real-scene 3D model with the support of selected tie points such as roofs and corner points. The transformation matrix was then calculated and matched to the coordinate systems of the two models. Finally, the IFC format of the sub-BIMs was converted into the b3dm format, and the two models were fused in the physical format.

Figs. 11–13 show the results of the fusion of the BIM and real-scene 3D model of the Old Hall. In the fused model, the real-scene 3D model and BIM were placed at the same location in space, similar to a single model, as shown in Fig. 13. As the "skin" of the BIM, the real-scene 3D model "decorates" the BIM model to describe the external situation of the Old Hall of Nanjing Museum. However, the real-scene 3D model controls the spatial positioning of the BIM. On the other hand, the BIM provides a skeleton for the real-scene 3D model, thus laying a basis for locating health monitoring sensors, which makes it possible to monitor

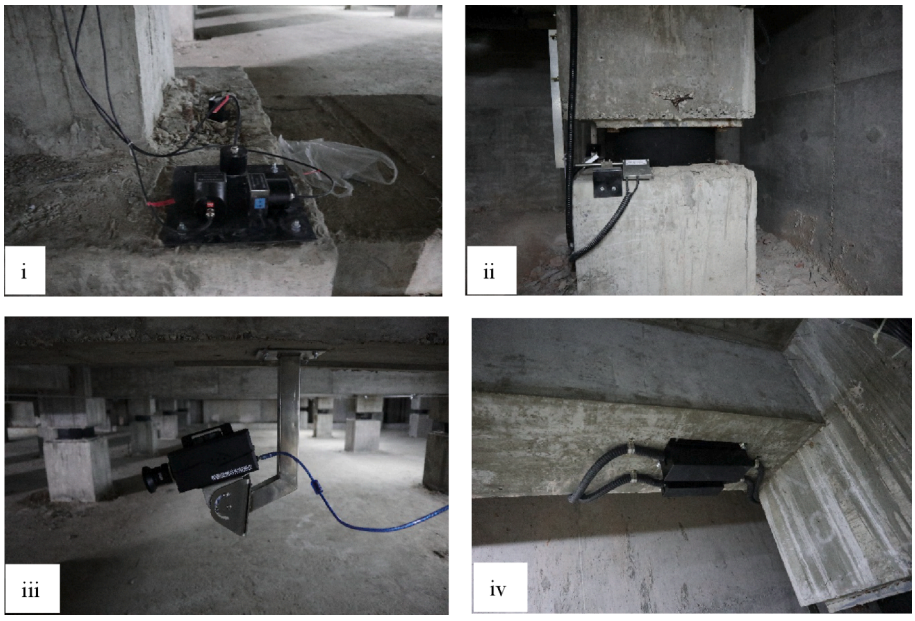


Fig. 9. Illustration of installed sensors. (i) Vibration sensor at location A; (ii) displacement sensor (optical fiber) at location C; (iii) displacement sensor (computer vision) at location A; (iv) temperature/strain sensor at location H.

Table 2
Parameters of unpiloted aerial vehicle and carried sensors.

UAV type	Fly time	Sensor	Sensor size	Focal length	Pix	Resolution
DJI M600	25 min	Sony Dsx-qx100	13.2 mm	10.4 mm	20.2 million	5472 × 3648 px

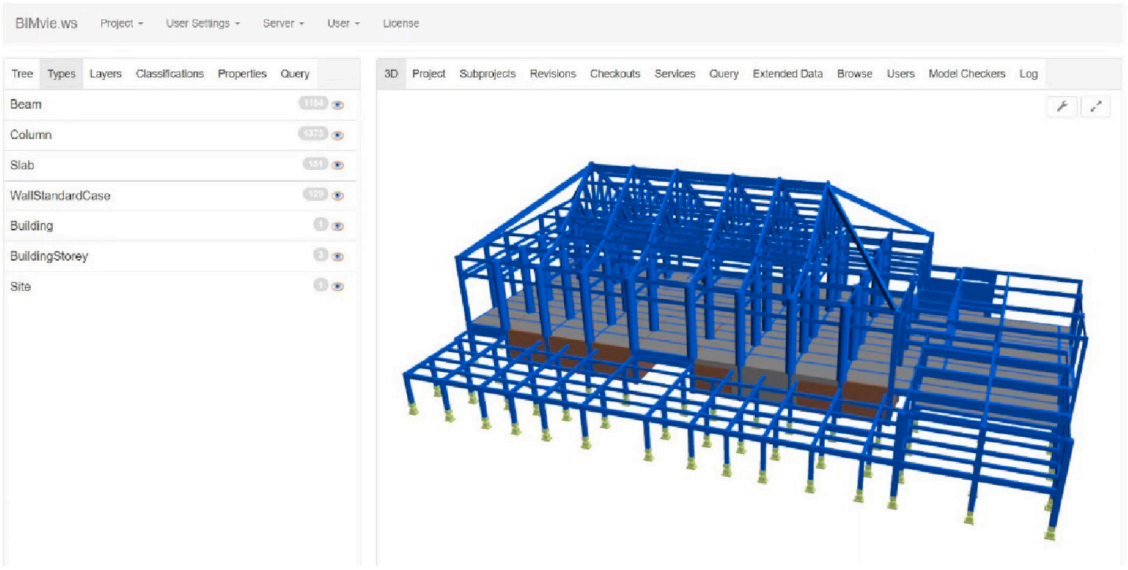


Fig. 10. Building information model of the Old Hall in BIM server before fusion.

Table 3
Component information of building information model of Old Hall of Nanjing Museum.

Types	Object number	Data volume of OBJ	Data volume of JSON
Ifcwalls	129	604 KB	43.2 MB
Ifcslabs	151	3.74 MB	47.9 MB
Ifcbeams	1154	6.08 MB	443 MB
Ifccolumns	1373	516 KB	421 MB
Ifcbuildingelementproxy	56	1.27 MB	2.50 MB

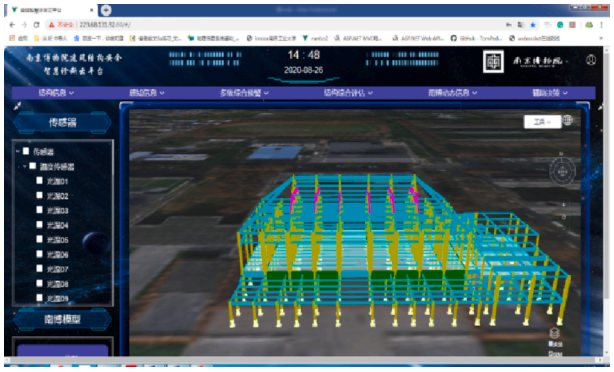


Fig. 11. BIM of the Old Hall in 3D Tiles format after fusion.

the details of a building's structural health. For example, vibration sensors are deployed on the key beams and columns on different building floors, thus supporting the structural and mechanical analysis of the building and facilitating judgments about the building's health state. Fig. 11 shows the BIM of the Old Hall after conversion from the IFC to the 3D Tiles standard. For convenience in displaying the fusion results, Fig. 12 is deliberately separated into two figures: the left side shows the real-scene 3D model of the Old Hall, and the right side shows the BIM behind the real-scene 3D model. Fig. 13 shows the fused model, in which the two models are in the same spatial position on the map and have been fused into the same software system.

Under the support of the fused model, the building SHM system for the Old Hall was developed, as shown in Fig. 13. The system adopts the Browser/Server architecture: the server side is developed with Visual

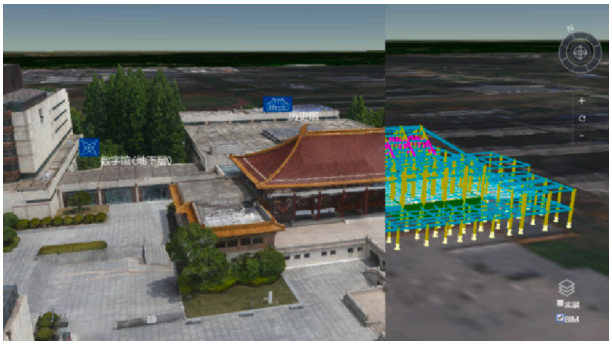


Fig. 12. The fusion of BIM and the real-scene 3D model.

Studio C#, the client side is developed with Vue.js, and the 3D engine on the client side is Cesium. A Microsoft SQL server was used to store the monitoring data and table structure, as shown in Table 4. Fig. 13 shows the six main functions that were developed: (1) The sensor information function manages the design drawings, BIM, and real-scene 3D model. (2) The early warning function manages sensor parameter information and real-time monitoring data. (3) The structural safety evaluation function provides early warnings for building safety according to real-time monitoring data from sensors and preset thresholds. However, it is a reserved function and will be developed in the future. (4) The Nanjing Museum information function introduces (5) the Nanjing Museum's auxiliary decision-making function. Preplans of the Nanjing Museum were stored and managed using this function. (6) The structural information function manages information regarding the building structure.

This study focused on integrating monitoring data and a fused 3D model. The Dev_id field name connects and integrates monitoring data with the corresponding sensor located on the fused model, as shown in Fig. 14. After clicking on the sensor icon in the SHM system (indicated by the red circle in Fig. 14), the real-time monitoring data of the selected sensor are displayed as shown in Fig. 15. In summary, integrating BIMs, real-scene 3D models, and monitoring data fosters the digital twin model of the Old Hall of the Nanjing Museum, which also supports building SHM for the Old Hall.

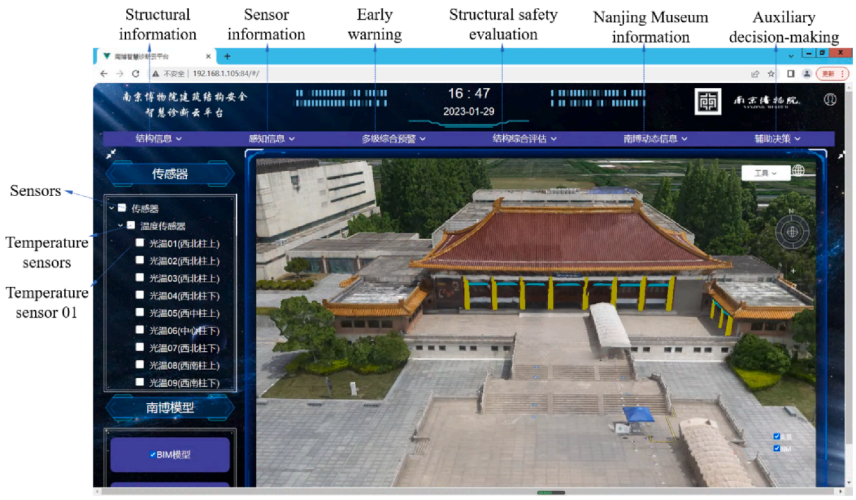


Fig. 13. Interface of the structural heath monitoring (SHM) system based on the fused model.

Table 4
Table structure of sensor data.

Field name	Data type	Description
ID	Int	Record No.
Dev_id	Vchar(10)	Sensor identification
Sensordata_starttime	Datetime	The time received data
Rawdata	Binary	Monitoring data of sensor

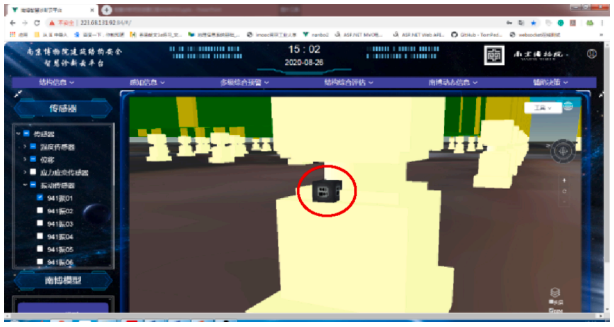


Fig. 14. Location of the sensor on the fusion model.



Fig. 15. Illustration of the sensor monitoring data.

4.2. Discussion

Monitoring a building's health status is complicated and establishing an effective building health-monitoring information model is important. By combining a BIM and a real-scene 3D model, we established an SHM information model based on digital twin technology. This information model supports the sensor arrangement, analysis of sensing data, and representation of a building's structural and mechanical status. The BIM provides a foundation for sensor location and mechanical state analysis of building structures. In contrast, the real-scene 3D model supports analyzing external environmental influences. The fused model integrates internal and external representations of a building. We propose that the fusion method can be applied to other fields, such as smart city planning, decision-making, building operations, and building maintenance.

The challenges of fusing BIMs and real-scene 3D models include different organizational methods regarding geometry, semantics, and space. The IFC standard for BIMs was selected as the source in the fusion method. The 3D Tiles standard for the real-scene 3D model was selected as the target. Both open-source standards are convenient for converting and supporting related model standards. Semantic information is extracted before geometric transformation and restored after conversion. Therefore, the semantic information of a BIM can be retained in the fusion process, which is essential for applying the fused model and constructing the digital twin. Following open standards, geometric and semantic information fusion adopts open-source software: a BIM server and Cesium library. This will assist in the promotion and application of the fusion method.

The spatial positional information of the building was fused using a 3D registration method. Based on the tie-point pairs of the two models, we used the rigid-body transformation method to determine the transformation relationship between the two coordinate systems. The transformation matrix is decomposed into several basic operations, namely, rotation and translation, using the SVD method. This makes the complex transformation easy to perform and realize.

Considering the Old Hall of the Nanjing Museum as a case study, a fusion model was developed. The merging process between the BIM and real-scene 3D model can be automatically accomplished using a program, excluding the tie-point selection. This was performed manually in this study. The automatic transformation process consumes approximately 25 min on a standard laptop (CPU is an Intel Core i7-10510U at 2.30 GHz). In this case, the programs for each merging step are separate. In the future, we can unify all the programs and select the tie points as input conditions to automate and accelerate the entire merging process.

The fused model results show that the buildings' geometric graphics and semantic information are preserved after fusion. Furthermore, the spatial positions of the two models are consistent with the external and internal scenarios of the building display. Thus, the micro-internal details of the building can be shown in a 3D macro space, and a 3D information model with a realistic scene and rich information can be built.

Based on the fused model of the Old Hall, a building SHM system was developed, in which 60 sensors for vibration, temperature, stress, and strain were arranged. The monitoring data were collected in real-time and integrated with the fused model, which formed the digital twin model for building health. The application to the Old Hall verifies the effectiveness of the BIM and real-scene 3D model fusion method that establishes a building health monitoring information model for a digital twin of the building.

Although we have achieved the integration of a BIM and a real-scene 3D model in terms of geometry, semantics, spatialization, and expression standards, a few unresolved issues remain. To effectively speed up the display of a real-scene 3D model, the level of detail method is typically used to establish different detail levels of the model. However, the level of detail method of the BIM is not established in the current fusion process, which decreases its loading speed. Furthermore, the BIM of a building may include numerous detailed components and large volumes of data, which may not be required to complete the process. Thus, simplifying and reducing the BIM is a problem worthy of further exploration. In addition, effectively analyzing the SHM data collected by all the arranged sensors is a challenge. The digital twin model can connect monitoring data from a spatial angle that offers a synthetic perspective to analyze the data, which needs further exploration.

5. Conclusion

Building safety is an important aspect of urban safety and disaster reduction. We propose a digital-twin-oriented building health monitoring information model. The model fuses a BIM with a real-scene 3D model, achieves the integrated management of a building's internal components and external environment, and combines a building's spatial shape and state. The greatest challenge in establishing a digital model is the fusion of geometric, semantic, and spatial location information between the BIM and real-scene 3D model. This work was completed with the support of multiple open-source standards and software, such as IFC, 3D Tiles, OBJ, and BIM server, which make the fusion method easier to use. The fusion process applied a registration method based on rigid-body transformation and a BIM semantic transformation method based on JSON.

The fused model integrates the monitoring data by providing a spatial description of the monitoring sensors. The monitoring data of the sensors distributed in the local part of the building can be connected in space, thus providing a foundation and support for analyzing the health status of the building from the perspective of the entire building. Thus, the response status of the building structure under the influence of the external environment can be dynamically represented, such as through the stress, strain, and vibration of the components. Considering Nanjing Museum's Old Hall as an example, we established a digital twin model and developed a building safety monitoring system based on this model, which preliminary verified the accuracy of our digital twin model. However, dynamically analyzing the overall health status of buildings according to monitoring data obtained locally requires further investigation. The proposed digital twin model can provide an overall perspective and an integrated data management platform for different monitoring data from sensors arranged in a building. Future studies should focus on examining the dynamic inversion and real-time presentation methods for the overall health status of buildings based on the digital twin model.

CRediT authorship contribution statement

Jinghai Xu: Conceptualization, Methodology, Writing- Original draft preparation. **Xin Shu:** Case study. **Peng Qiao:** Case study. **Shanshan Li:** Writing- Reviewing and Editing. **Jigang Xu:** Case study.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

This work was jointly supported by The National Natural Science Foundation of China (Grant No. 42271420), the Science and Technology Plan Project of the Ministry of Housing and Urban-Rural Development of China in 2022 (Grant No. 2022-K-041). The authors would like to express their appreciation to the editor and the referees for their comments and suggestions that greatly improved the manuscript.

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